

Effects of fish meal replacement by meat and bone meal supplemented with garlic (*Allium sativum*) powder on biological indices, feeding, muscle composition, fatty acid and amino acid profiles of whiteleg shrimp (*Litopenaeus vannamei*)

Taghi Tazikheh¹ | Abdolmohammad Abedian Kenari¹  | Noah Esmaeili²

¹Department of Aquaculture, Faculty of Natural resources and Marine Science, Tarbiat Modares University, Noor, Iran

²Institute for Marine and Antarctic Studies, University of Tasmania, Hobart, Tasmania, Australia

Correspondence

Abdolmohammad Abedian Kenari,
Department of Aquaculture, Faculty of Natural Resources and Marine Sciences,
Tarbiat Modares University, PO Box 64414-356, Noor, Mazandaran, Iran.
Emails: aabedian@modares.ac.ir; aabedian@yahoo.co.uk

Funding information

Tarbiat Modares University

Abstract

Shortage of fish meal (FM) resources in the marine environment, high price and high demand for this ingredient led researchers to seek for alternative protein resources such as meat and bone meal (MBM). The effects of adding MBM and garlic powder to the shrimp diet on growth performance, muscle composition, amino acid and fatty acid profiles of whiteleg shrimp (5.00 ± 0.23 g) were investigated. A total of six dietary treatments were formulated: 0%MBM (control without garlic powder), 25%MBM and 50%MBM (treatments without garlic powder; WG treatments); 0%MBM+G (control with garlic powder), 25%MBM+G and 50%MBM+G (treatments with garlic powder; G treatments). Our results indicated that increase MBM level in the diets (up to 50%MBM) impaired growth performance, muscle composition, fatty acid and amino acid profiles in the muscle of shrimp. Interestingly, garlic supplementation improved the aforementioned factors so that shrimp fed 50%MBM+G diet had no significant difference when compared to 0%MBM group. The current results indicated that garlic could increase levels of FM replacement by increasing feed intake, improving protein and amino acid utilization. Overall, 50%MBM+G diet is recommended for use in shrimp production.

KEYWORDS

aquaculture, crustacean, growth, herbal medicine, nutrition, omega 3

1 | INTRODUCTION

Providing food for more than nine billion people can be challenging by the middle of the twenty-first century, and impacts of climate change and environmental concerns can worsen this problem (FAO, 2018). Among different agriculture sectors, aquaculture can be the best candidate to produce food sustainably. In 2016, 88% of aquaculture production (total: 171 million tonnes and half of that for aquaculture by 598 species) were consumed at a per capita rate of 20.3 kg. This amount was the highest record due to relatively stable capture fisheries production, reduced wastage and continued aquaculture growth rate.

Furthermore, more than 60% of shrimp production is based on artificial feed, which costs 50%–60% of the total operating costs. In order to produce shrimp sustainably, reducing production costs is an overriding priority. Finding a sustainable replacement for the most expensive ingredients in diet such as FM and fish oil is the best option (Webster, Tiu, & Tidwell, 1997). According to FAO, 2018, fish meal and fish oil are still considered the most nutritious and digestible ingredients for farmed species feeds. However, their inclusion rates in diets have indicated a clear decreasing trend as they are performed more selectively and precisely. Therefore, many studies worldwide have taken an attention

on FM replacement with alternative protein sources such as insects and animal by-products particularly meat and bone meal (MBM). Animal protein sources are considered as potentially sustainable feed ingredients due to lower carbohydrate content, the high amount of protein and phosphorus, lower price, and acceptable amino acid profile (Meeker, 2006). However, lower digestibility in comparison with FM, lysine and methionine deficiency, and also high ash content are some restrictions of using MBM (Bureau, Harris, & Cho, 1999).

There are some studies that applied MBM in the diet of whiteleg shrimp (*Litopenaeus vannamei*) (Forster, Dominy, Obaldo, & Tacon, 2003; Hernández, Olvera-Novoa, Aguilar-Vejar, González-Rodríguez, & de la Parra, 2008; Tan et al., 2005) and giant freshwater prawn (*Macrobrachium rosenbergii*) (Arshad Hossain & Farzana Islam, 2007). Foster et al. (2003) tested three different MBM so that they could replace 75% of FM in shrimp diet by MBM (530 g/kg protein, 170 g/kg fat and 190 g/kg ash) without impairing growth performance. Also, Hernandez et al. (2008) reported that MBM (530 g/kg protein, 100 g/kg fat, 240 g/kg ash) is an alternative that can replace up to 35% of FM protein in whiteleg shrimp diets without significant decrease in growth performance, survival rate, feed conversion ratio (FCR), protein efficiency ratio (PER) and muscle composition. As we can see, there is a big discrepancy between studies probably due to the different composition of MBM. In this way, studying FM replacement by different MBM sources is necessary.

Growth and feed efficiency in shrimp species is mainly related to composition of the nutrients, but many factors have key roles in digestion, absorption and transfer of nutrients in the intestine. Accordingly, the use of some ingredients which stimulate digestive enzymes can improve digestibility and eventually improve fish growth performance. Herbal medicines, such as garlic (*Allium sativum*), are among these supplements. Garlic aids digestion and absorption, first by stimulating digestive enzymes like lipase, alkaline phosphatase and trypsin, which helps to improve feed digestion. Secondly, garlic stimulates bile acid that plays a key role in fatty acid digestibility and absorption (Platel & Srinivasan, 2004).

Garlic caused an increase in growth performance in fish such as sobaity sea bream (*Sparidentex hasta*) (Jahanjoo, Yahyavi, Akrami, & Bahri, 2018), rainbow trout (*Oncorhynchus mykiss*) (Büyükdeveci, Balcázar, Demirkale, & Dikel, 2018; Esmaeili, Abedian Kenari, & Rombenso, 2017; Esmaeili, Kenari, & Rombenso, 2017), sterlet sturgeon (*Acipenser ruthenus*) (Lee et al., 2014), Asian sea bass (*Lates calcarifer*) (Talpur & Ikhwannuddin, 2012), Nile Tilapia (*Oreochromis niloticus*) (Naqi et al., 2019), common carp (*Cyprinus carpio*) (Karimi Pashaki, Ghasemi, Sharif Rohani, & Hosseini, 2018) and goldfish (*Carassius auratus*) (Dadgar et al., 2019). Also, there are a few studies that reported the positive effects of garlic in whiteleg shrimp growth performance (Javadzadeh, Salarzadeh, Yahyavi, Hafezieh, & Darvishpour, 2012; Labrador, Guíñares, & Hontiveros, 2016). Garlic has many effects such as antibacterial and antifungal properties, improvement of meat quality, body metabolism, growth performance,

the immune system and the increase in feed consumption (Lee & Gao, 2012). It is noted that garlic contains allicin, which results in improved intestinal flora function, thereby improving digestion which ultimately leads to improved growth performance (Khalil, Nadia, & Soliman, 2001).

Whiteleg shrimp is distinctly at the top of the major species in the crustacean sector worldwide by 53% of total crustacean aquaculture production (7.862 million tonnes). In Iran, whiteleg shrimp is the main crustacean species and a complete cycle of production of this species has been completed. According to the Iran Fishery Organization, it is predicted that in 2020, the amount of whiteleg shrimp production would reach 60,000 tonnes per year. Accordingly, we designed this experiment to investigate whether dietary garlic exhibits a positive effect on increasing levels of FM replacement by MBM in whiteleg shrimp diet or not.

According to our knowledge, there is no study to investigate the combined effect of animal proteins and herbal medicine in crustacean performance. Also, most studies related to FM replacing by MBM are for last decades and need to be updated. In this way, we evaluated the growth performance, feeding rate, amino acid and fatty acid profiles of whiteleg shrimp fed diets including 0%, 148 g/kg (25% replacement) and 303 g/kg (50% replacement) MBM with g/kg or without garlic supplementation. This work is part of our project about the effect of herbal medicine on fish and shrimp physiology that some of outputs have already been published (Ramezanzadeh, Abedian Kenari, & Esmaeili, 2019; S. Ramezanzadeh, A. Abedian Kenari, M. Esmaeili, & A. Rombenso, unpublished data; Safavi, Abedian Kenari, Tabarsa, & Esmaeili, 2019; Zeilab Sendijani, Abedian Kenari, Smiley, & Esmaeili, 2019).

2 | MATERIALS AND METHODS

2.1 | Diet preparation

Six isonitrogenous (350 g/kg crude protein) and isoenergetic (17 MJ/Kg) diets were formulated to gain 0%, 25% and 50% of FM replacement by MBM (Table 1). Control diets with (0%MBM+G) or without (0%MBM) garlic contained 454 g/kg of FM (Kilka fish meal, Pars Kilka Company). The experimental diets were based on 0%MBM+G and 0%MBM, followed by increasing level of MBM (Babolsar Company) according to FM replacement (25% and 50%) with (G) or without (WG) garlic: 25%MBM+G and 50%MBM+G (G treatments: with garlic powder), 25%MBM and 50%MBM (WG treatments: without garlic powder).

2.2 | Whiteleg shrimp and experimental condition

The whiteleg shrimp (average weight of 5 ± 0.23 g) were farmed at the Center of Reproduction and Cultivation of whiteleg shrimp (Gomishan, Golestan, Iran). After acclimation for two weeks, they were transferred to 18 fibreglass tanks (six treatments in

TABLE 1 Formulation and proximate composition of experimental diets fed to whiteleg shrimp during 60 days

Ingredients	Without garlic			With garlic		
	0% MBM	25% MBM	50% MBM	0% MBM+G	25% MBM+G	50% MBM+G
g/kg, as fed basis						
Fish meal	454	340.5	227	454	340.5	227
MBM	0	148.1	303.4	0	148.1	303.4
Wheat flour	422.1	408.6	366.8	422.1	408.6	366.8
Fish oil	5	5	5	5	5	5
Soy oil	5	5	5	5	5	5
Lecithin	5	5	5	5	5	5
Dicalcium phosphate	5	5	5	5	5	5
Mineral ^a	25	25	25	25	25	25
Vitamin supplement ^b	15	15	15	15	15	15
Antifungal	2.5	2.5	2.5	2.5	2.5	2.5
Antioxidant	0.2	0.2	0.2	0.2	0.2	0.2
Lysine	0	5	5	0	5	5
Methionine	0	5	5	0	5	5
CMC	61.17	30	30	31.17	0	0
Garlic powder	0	0	0	30	30	30
Proximate composition (wet basis)						
Protein (%)	35.42	34.92	35.62	34.71	35.04	35.28
Fat (%)	8.18	8.63	8.30	8.21	8.58	8.40
Ash (%)	14.66	15.29	15.01	15.75	15.51	15.47
Carbohydrate (%)	32.78	32.50	32	31.93	32.50	31.95
Moisture (%)	8.96	8.66	9.14	9.1	8.37	8.4
Energy (kJ/g) ^c	17.17	17.23	17.26	17.61	17.24	17.33

Note: Dietary treatments with different levels of fish meal (FM) replacement with meat and bone meal (MBM) with (G) or without (WG) garlic powder: 0%MBM (control without garlic powder), 25%MBM and 50%MBM (treatments without garlic powder; WG treatments). 0%MBM+G (control with garlic powder), 25%MBM+G and 50%MBM+G (treatments with garlic powder; G treatments).

^a1 kg mineral supplementation contained: Fe (6 g), Cu (600 mg), Mn (5 g), Zn (10 g), I (600 mg), Se (20 mg), Co(100 mg), choline chloride (6 g), Ca (10,000 mg)

^b1 kg vitamin supplementation 0.5% contained: A (1,200,000 IU), D₃(400,000 IU), E (50,000 m), K₃ (800 mg), niacin (35,000 mg), B₉ (1,000 mg), C (2,500 mg), B₁ (30,000 mg), B₂ (4,000 mg), B₆(25,000 mg), inositol (50,000 mg), biotin (150 mg).

^cCalculated on the basis of 23.6, 39.5 and 17.2 KJ/g of protein, lipid and carbohydrate respectively (NRC, 2011).

triplicate) of 300 L for the experiment composed by a semi-re-circulation system, with a stocking density of 25 whiteleg shrimp per tank. Before feeding, we removed faeces, uneaten feed and other materials by siphoning tanks every morning and replaced 20%-30% of the water. In order to supply oxygen, the used air rock was connected to an aerated pump. Water quality parameters were regularly monitored during the experimental period. Measurement of water temperature (mercury thermometer, Zomorodazma Company), dissolved oxygen (CyberScan, Eutech Instruments, DO 110), was done daily, and total ammonium was lower than 0.05 mg/L during the 60-day experimental period. Salinity levels (Refractometer ATAGO) and pH (Hanna instrument, 8314) were measured weekly (Hosseinpour Aghaei, Abedian Kenari, Yazdani Sadati, & Esmaeili, 2018). Measured

factors ranged 27–29°C, 6.2–7.5 mg/L O₂, 29–30 g/L salinity and pH 6.8 – 7.5 respectively. The experiment was carried out in a covered room with a light period of 12 hr of light and 12 hr of darkness. Whiteleg shrimp were hand-fed to apparent satiation three times daily.

2.3 | Measurements of growth parameters

Weighing the whiteleg shrimp with a precision scale of 0.01 g was done every 20 days. After the experimental period (60 days), the whiteleg shrimp was washed with fresh water after 24 hr of starvation (in order to empty the gastrointestinal tract) and fresh salt was removed from the body. All growth formulas are reported in

TABLE 2 Fatty acid profile of experimental diets was fed to whiteleg shrimp with replacing fish meal by meat and bone meal during 60 days (g/100 g lipid)

	Key feed ingredient		Without garlic			With garlic			
Fatty acids	FM	MBM	0% MBM	25% MBM	50% MBM	0% MBM+G	25% MBM+G	50% MBM+G	SDM
g/100 g fatty acid methyl esters (FAMES)									
C14	3.12	0.81	1.22	1.45	1.03	1.50	1.64	1.77	0.21
C16	20.21	24.02	19.89	20.19	21.12	19.86	20.28	21.15	1.85
C18	5.43	6.55	4.02	3.54	3.22	3.86	3.80	4.43	0.69
C20	1.28	1.21	3.21	3.52	3.99	3.45	3.75	4.00	0.42
SFAs ^a	31.04	33.44	29.63	29.31	30.10	29.56	30.38	31.75	2.09
C16:1	4.21	6.34	4.08	4.45	5.21	4.43	4.76	5.13	0.60
C18:1n9	26.32	41.12	25.30	27.26	28.62	25.49	27.92	28.78	2.12
MUFAs ^b	30.53	47.46	29.38	31.71	33.74	29.92	32.68	33.91	2.15
C18:2n6	3.67	13.75	16.52	17.60	20.08	16.65	18.59	19.44	1.11
C18:3n6	0.33	0.26	2.60	2.66	2.14	2.43	2.63	2.12	0.42
C20:4n6	0.17	0.22	2.22	2.46	1.40	2.58	1.90	1.69	0.21
C18:3n3	1.36	0.59	0.84	0.79	0.67	0.91	0.79	1.19	0.14
C20:5n3	5.12	0.30	3.60	2.87	2.16	2.70	2.57	2.87	0.32
C22:6n3	18.04	0.54	9.46	7.61	5.79	9.37	7.22	4.66	0.54
PUFAs ^c	28.69	15.66	35.24	33.99	32.24	34.64	33.70	31.97	1.94
LC-PUFA ^d	40.25	1.09	15.48	12.94	10.77	14.69	11.88	9.40	1.96
Σn3 ^e	24.52	1.43	13.90	11.27	8.62	12.98	10.58	8.72	1.55
Σn6 ^f	4.17	14.23	19.34	22.86	23.62	21.66	23.12	23.25	0.96
n3/n6	5.88	0.10	0.71	0.49	0.36	0.59	0.46	0.37	0.04

Note: Dietary treatments with different levels of fish meal (FM) replacement with meat and bone meal (MBM) with (G) or without (WG) garlic powder: 0%MBM (control without garlic powder), 25%MBM and 50%MBM (treatments without garlic powder; WG treatments). 0%MBM+G (control with garlic powder), 25%MBM+G and 50%MBM+G (treatments with garlic powder; G treatments).

^aSaturated fatty acids—sum of all fatty acids without double bonds; includes 17:0, and 22:0 in addition to individually reported SFAs.

^bMonounsaturated fatty acids—sum of all fatty acids with a single double bond; includes 17:1n-7, 22:1n-11 and 22:1n-9 in addition to individually reported MUFAs.

^cPolyunsaturated fatty acids—sum of all fatty acids with ≥2 double bonds; includes 16:2n-4 in addition to individually reported PUFAs.

^dLong-chain polyunsaturated fatty acids; sum of all C20 and C22 fatty acids with three or more double bonds; includes 20:3(n-6) in addition to other individually reported LC-PUFAs.

^eSum of all n-3 fatty acids.

^fSum of all n-6 fatty acids; includes 20:3n-6 in addition to individually reported n-6 fatty acids.

the footnotes of the Table 2 according our previous works (Kenari, Sotoudeh, & Rezaei, 2011; Oujifard, Seyfabadi, Kenari, & Rezaei, 2012). For analysing muscle composition, fatty acid and amino acid profiles, 10 individual whiteleg shrimp were randomly selected from each replicate and then placed in a -20°C freezer.

Garlic powder was added to diets at 30 g/kg level according to optimum level previously reported in whiteleg shrimp works (Labrador et al., 2016). The garlic was purchased from the local market, and after hulling, hashing and drying in an oven at 50°C for 48 hr, the dried garlic was ground by mill (Moulinex AR1066Q). FM, MBM and other ingredients were completely mixed using a mixer (Philips HR7628). Then, we added kilka fish oil (Pars Kilka company), soybean oil (Kesht Va Sanat Shomal Company), and tepid water and mixed properly again. In the next step, the paste was pelleted using a meat mincer (CGT Company, 2mec, Italy) with

1mm die and then dried in oven by 50°C for 18-24hrs to > 90% dry matter. Finally, after packing prepared diets in nylon bags, they stored in -20°C.

2.4 | Chemical analysis

To assess the chemical composition of muscle and diets including moisture, ash, crude fat and crude protein contents, the method of AOAC was used (AOAC, 1995). The moisture content was obtained from the weight difference from the specimens for 12-24 hr in an oven at 105°C. To determine the ash content, 0.5 g of samples were burned in a furnace at 550°C for 5 hr. The Kjeldahl method was used to measure the crude protein content using a Kjeltex Auto Analyzer (Kjeltex Analyzer Unit 2300) with a conversion factor of 6.25. The amount of lipid was measured by

Soxtec method (Soxtec 2050 FOSS Model). Nitrogen-free extract (NFE) plus fibre expressed as carbohydrate was calculated by the difference:

$$100 - (\text{Protein} + \text{fat} + \text{ash} + \text{moisture}) \text{ (Aksnes \& Opstvedt, 1998).}$$

The gross energy of the diet and faeces was calculated according to below formula (NRC, 2011):

$$((\text{Energy (MJ/kg)} = (\text{Protein} \times 23.6 \text{ Kj/g}) + (\text{Fat} \times 39.5 \text{ Kj/g}) + (\text{Carbohydrate} \times 17.2 \text{ Kj/g})).$$

2.5 | Amino acid analyses

For analysing of amino acids, method of o-phthalaldehyde (OPA) was used (Oujifard et al., 2012). Briefly, after hydrolysis of samples in 6 N HCl for 24 hr at 110°C, samples were derivatized with OPA. The total amino acids were analysed by HPLC (Knauer) using C18 column (Knauer) at the flow rate of 1 ml/min with fluorescence detector (RF-530; Knauer). Also, requirement of whiteleg shrimp for amino acids (Akiyama, 1992) is reported in Table 3.

2.6 | Fatty acid analyses

For fatty acid analysis, the extraction of total lipid of diets and the muscle of whiteleg shrimp were done by chloroform/methanol method (Folch, Lees, & Sloane-Stanley, 1957). Following adding methanol to the sample, lipid was methanolized with boron trifluoride (BF₃). After observing of the two phases, the upper layer was separated and kept in -20°C until infusion to GC mass instrument (Firestone, 2009). From GC (Varian Analytical Instrument, CP 3800) equipped with a flame ionization detector fitted with a permanently bonded polyethylene glycol, fused silica capillary column (PBX70 SGE Analytical Science; 120 m × 0.25 mm internal diameter, film thickness 0.25 µm, Melbourne, Australia) was used for separating methyl esters (FAMES) fatty acids. The injection volume was 1.0 µl, and the carrier gas was helium. The injector and detector temperature was, respectively, 230°C and 260°C. A split injection approach of 20:1 was used, and the temperature was programmed

TABLE 3 Amino acid profile of experimental diets was fed to whiteleg shrimp with replacing fish meal by meat and bone meal during 60 days (g/100 g)

	Key feed ingredient		Without garlic			With garlic			R ^a	SDM
	FM	MBM	0%	25%	50%	0%	25%	50%		
			MBM	MBM	MBM	MBM+G	MBM+G	MBM+G		
Essential amino acids (EAAs)% fed basis for shrimp										
Arginine	3.64	3.99	2.75	3.62	3.29	3.86	3.59	3.31	2.20	0.32
Histidine	1.71	1.02	1.20	1.49	1.08	1.24	1.56	1.05	0.80	0.15
Isoleucine	2.38	1.57	1.45	1.65	1.25	1.52	1.52	1.25	1.30	0.15
Leucine	3.68	3.43	3.41	3.61	3.22	3.61	3.64	3.27	2.10	0.35
Lysine	4.84	2.60	3.43	3.13	2.85	3.41	3.05	2.68	2.00	0.34
Methionine	1.45	0.51	1.32	1.53	1.11	1.49	1.48	1.14	1.40	0.14
Phenylalanine	2.55	2.11	2.75	2.04	2.61	2.06	2.27	2.20	2.70	0.13
Threonine	2.32	1.77	1.58	1.73	1.36	1.84	1.67	1.35	1.40	0.21
Valine	2.99	2.50	2.26	2.51	1.97	2.27	2.50	1.94	1.50	0.38
Non-essential amino acids % fed basis for shrimp										
Alanine	4.04	3.55	3.16	3.46	2.97	3.24	3.55	2.94		0.40
Aspartic acid	6.15	4.12	4.71	4.97	4.37	4.01	4.87	4.55		0.47
Glutamic acid	8.33	7.15	7.37	7.75	6.92	7.25	7.70	7.11		0.51
Glycine	2.8	6.07	2.64	3.81	3.49	2.66	3.78	3.48		0.38
Serine	2.59	2.33	1.98	1.81	1.49	2.11	1.85	1.51		0.25
Tyrosine	1.52	2.01	2.63	2.52	2.14	2.37	2.40	2.14		0.29
ΣEAA	25.56	19.5	20.15	21.13	18.74	21.30	21.28	18.19		1.25
ΣNEAA	25.43	25.23	22.49	24.32	21.38	21.64	24.15	21.73		1.50
EAA/ NEAA	1.00	0.77	0.89	0.86	0.87	0.98	0.88	0.84		0.13
Total amino acids	50.99	44.73	42.64	45.45	40.12	42.94	45.43	39.92		1.77

Note: Dietary treatments with different levels of fish meal (FM) replacement with meat and bone meal (MBM) with (G) or without (WG) garlic powder: 0%MBM (control without garlic powder), 25%MBM and 50%MBM (treatments without garlic powder; WG treatments). 0%MBM+G (control with garlic powder), 25%MBM+G and 50%MBM+G (treatments with garlic powder; G treatments).

Abbreviations: ΣEAA: sum of essential amino acids, ΣNEAA: sum of non-essential amino acids, TAAs: total amino acids.

^aRecommended requirement for shrimp (Akiyama, 1992).

to increase from 160°C to 180°C at a rate of 2°C/min and held at 180°C for 85 min. Individual FAMES were identified using the external standard as reference (Sigma-Aldrich) (Abtahi, Yousefi, & Kenari, 2013).

2.7 | Statistical analysis

Normality and homogeneity of variances of data have been checked by Kolmogorov-Smirnov test and Bartlett's test respectively. All data related to growth performance, muscle composition, fatty acid and amino acid profiles were analysed by one-way analysis of variance (ANOVA) using SPSS software (release 19.0 for Windows). Duncan's multiple comparisons test was performed to illustrate the differences between the treatment means. *P* values in the present study were in the level of *p* < .05.

3 | RESULTS

3.1 | Growth parameters

The results of growth indices and survival rate of whiteleg shrimps in response to different levels of MBM with and without garlic powder are shown in Table 4. According to this table, the mean of final weight in 50%MBM group (11.13 g) was significantly higher than the other treatments (*p* < .05). In this study, there was no significant difference in FCR, specific growth ratio (SGR) and survival ratio among groups but 25%MBM (1.23) had the lowest value of FCR. Usually, in groups without garlic SGR and FCR levels in shrimps fed garlic was higher and

lower than those fed dietary without garlic respectively. PER (1.70) and LER (7.17) values in shrimp fed 50%MBM diets were significantly lower than other groups, indicating the negative effect of MBM on protein and lipid efficiency. Food consumption (FC) in shrimp fed dietary supplemented with garlic was higher than treatments without garlic; and garlic could stimulate appetite of whiteleg shrimps.

3.2 | Approximate composition

According to Table 5, there was no significant difference in protein and fat contents in muscle among treatments. The higher amount of ash contents in shrimp fed 0%MBM (5.10%) and 25%MBM (5.81%) diets was observed when compared with those fed other groups (*p* < .05). Also, individuals fed diets with 50% replacing FM by MBM either with or without garlic had statistically higher amount of moisture as compared to those fed 25%MBM (74.50%) diet (*p* < .05). Generally, lipid and proteins, two important factors of chemical composition, were not affected by MBM or garlic supplementation in the diets.

3.3 | Amino acid profile

Table 2 shows amino acid profile of FM, MBM and experimental diets were fed to whiteleg shrimp during 60 days (g/100 g). Sum of essential amino acids (Σ EAA) and total amino acids (TAAs) in 50%MBM and 50%MBM+G diets were lower than others. In this sense, threonine, phenylalanine and methionine in 50%MBM (1.36, 2.01 and 1.11, respectively) and 50%MBM+G (1.35, 2.20

TABLE 4 Growth performance of whiteleg shrimp fed with different levels of meat and bone meal with adding garlic powder for 60 days. Values are represented means $\pm$ standard deviations. Means without letter are not significantly different. Letters a and b indicate significant differences in treatments, according to Duncan's multiple range tests (*p* < .05)

Growth indices	Without garlic			With garlic			SDM
	0% MBM	25%MBM	50%MBM	0%MBM+G	25% MBM+G	50%MBM+G	
Initial weight (g)	5.21	4.98	5.42	5.32	5.31	5.14	0.23
Final weight (g)	12.90 a	12.07 a	11.13 b	12.65 a	12.69 a	11.96 a	1.11
Weight gain (g)	7.69 a	7.09 a	5.71 b	7.33 a	7.38 a	6.82 a	0.57
FCR	1.41	1.27	1.79	1.53	1.48	1.72	0.23
SGR	4.03	3.91	3.74	3.99	3.98	3.89	0.62
PER	2.00 a	2.26 a	1.77 b	1.88 a	1.93 a	1.70 b	0.33
LER	8.73 a	9.20 a	7.61 b	8.23 ab	7.93 ab	7.17 b	0.75
FC	10.84 a	9.00 b	9.27 b	11.21 a	10.92 a	11.38 a	0.94
Survival (%)	85.33	85.33	93.33	93.33	94.66	85.33	0.33

Note: Dietary treatments with different levels of fish meal (FM) replacement with meat and bone meal (MBM) with (G) or without (WG) garlic powder: 0%MBM (control without garlic powder), 25%MBM and 50%MBM (treatments without garlic powder; WG treatments). 0%MBM+G (control with garlic powder), 25%MBM+G and 50%MBM+G (treatments with garlic powder; G treatments).

Abbreviations: Weight gain = (final weight – initial weight); FCR: feed conversion ratio = total dry feed intake (g)/wet weight gain (g); SGR: specific growth ratio (% body weight gain/day) = ((Ln final weight – Ln initial weight)/(experiment period)) $\times$ 100; PER: protein efficiency ratio = weight gain in g/protein intake in g; LER: lipid efficiency ratio = weight gain in g/lipid intake in g; FC: feed consumption = feed consumption for each shrimp; Survival% = (final number of shrimp/initial number of shrimp) $\times$ 100.

TABLE 5 The effect of replacing fish meal with meat and bone meal and garlic powder component on analysis of whiteleg shrimp muscle (100% wet weight). Values are represented means $\pm$ standard deviations. Means without letter are not significantly different. Letters a and b indicate significant differences in treatments, according to Duncan's multiple range tests ($p < .05$)

Analysis proximate	Without garlic			With garlic			SDM
	0%MBM	25%MBM	50%MBM	0%MBM+G	25%MBM+G	50%MBM+G	
Crude protein (%)	16.40	16.03	16.04	16.76	16.81	16.14	1.03
Crude lipid (%)	3.40	3.66	3.76	3.60	3.80	3.82	0.44
Ash (%)	5.10 a	5.81 a	4.63 b	4.45 b	4.40 b	4.14 b	0.76
Moisture (%)	75.19 ab	74.50 b	75.57 a	74.83 ab	74.99 ab	75.90 a	1.79

Note: Dietary treatments with different levels of fish meal (FM) replacement with meat and bone meal (MBM) with (G) or without (WG) garlic powder: 0%MBM (control without garlic powder), 25%MBM and 50%MBM (treatments without garlic powder; WG treatments). 0%MBM+G (control with garlic powder), 25%MBM+G and 50%MBM+G (treatments with garlic powder; G treatments).

and 1.14, respectively) diets were lower than requirements of whiteleg shrimp for these amino acids. The results of the amino acid values of whiteleg shrimp muscle are presented in Table 6. According to this table, TAA contents were not significantly different among treatments. With focus on each amino acid, arginine in whiteleg shrimp fed 50%MBM (7.20) and 50%MBM+G (7.63) had significantly lower value than the others ($p < .05$). Also, individuals fed 25%MBM diet had a higher amount of valine (3.99) in muscle as compared to whiteleg shrimp fed dietary 50%MBM (2.79) and 50%MBM+G (2.92) ($p < .05$). Whiteleg shrimp fed dietary garlic and especially 50%MBM+G diet had higher value of aspartic acid (9.04) when compared with those fed diets without garlic ($p < .05$). According to the results of ANOVA, TAA in individuals fed dietary garlic was higher than those fed diets without garlic so that shrimps fed 50%MBM and control diets had a lesser amount than the other groups ($p < .05$). Generally, garlic could improve amino acid metabolism in whiteleg shrimp and although 50%MBM+G diet had lower TAA (39.92) as compared to other diets, whiteleg shrimp fed this diet had higher value of TAA in muscle (78.20).

3.4 | Fatty acid profiles

Fatty acid profiles of experimental diets are presented in the Table 3. As observed, the amount of polyunsaturated fatty acids (PUFAs) and sum of n-3 fatty acids ($\Sigma n3$) decreased with increasing the level of MBM in the diets. On the other hand, the amount of monounsaturated fatty acids (MUFAs) decreased with increasing the level of MBM in the diets. The results of the fatty acid profiles in the muscle of whiteleg shrimp are presented in Table 7. The amount of saturated fatty acids (SFAs) in those fed 25%MBM+G diet (27.90) was significantly lower than shrimp fed control (29.56) and 0%MBM+G (29.97) diets ($p < .05$). Furthermore, C18:3n6 in the muscle of the whiteleg shrimp fed control group (0.72) had lesser contents than the others ($p < .05$). DHA (C22:6n-3) in the presented study did change statistically between whiteleg shrimps fed control and 50%MBM+G diets. However, individuals fed 50%MBM diet had lower levels of DHA (11.52) compared to other groups. In the present study, PUFA

and $\Sigma n3$ in the muscle of whiteleg shrimp fed 50%MBM (35.86, 21.19) were significantly lower than whiteleg shrimp fed dietary 0%MBM+G (39.54, 24.1) ($p < .05$). Also, LC-PUFA experienced similar trend so that shrimp fed 50%MBM diet (23.39) had lower value when compared with control groups ($p < .05$). All in all, both replacing FM by MBM and garlic had a significant effect on the amount of PUFA and $\Sigma n3$ in muscle.

4 | DISCUSSION

4.1 | Growth parameters

As the amount of FM in the commercial diet of whiteleg shrimp is high (150–250 g/kg (Cao et al., 2015; Sookying, Davis, & Soller Dias Da Silva, 2013; Turchini, Trushenski, & Glencross, 2019) and also this species is the most farmed shrimp worldwide; whiteleg shrimp are among the largest consumers of fishmeal (Olsen & Hasan, 2012). In this way, the replacement of FM by other protein sources is an important action towards the sustainability of whiteleg shrimp production. Among potential alternative protein sources, MBM has been suitably formulated in whiteleg shrimp diets (Forster et al., 2003; Hernández et al., 2008; Tan et al., 2005) and giant freshwater prawn (Arshad Hossain & Farzana Islam, 2007). Recently, the amount of fishmeal in whiteleg shrimp diet could be declined from 250 g/kg to 100 g/kg without affecting the growth performance (Xie, Niu, Zhou, Liu, & Tian, 2018). The results of our study indicated that the use of MBM up to 303 g/kg and 30 g/kg garlic supplementation in the diet of whiteleg shrimp without decreasing growth performance is possible. However, from a review of the literature surrounding fish and shrimp, we found that the level of replacing is strongly related to the amount of FM in control diet and also MBM chemical composition. In the present study, we formulated 450 g/kg FM in diet which is higher than the amount of FM in commercial shrimp diet in Iran (up to 300 g/kg). We considered this optimum level as a control diet and then we investigated whiteleg shrimp response to the FM replacement by MBM. Our results indicated at least 340 g/kg FM in whiteleg shrimp diet (25% FM replacement by MBM) is necessary

TABLE 6 Effect of fish meal replacement by meat and bone meal on amino acid profile of whiteleg shrimp muscle fed on various treatments after 60-day (g/100 g). Values are represented means $\pm$ standard deviations of triplicate samples. Means without letter are not significantly different. Letters a and b indicate significant differences in treatments, according to Duncan's multiple range tests ($p < .05$)

% (EAA)	Without garlic			With garlic			SDM
	0%MBM	25%MBM	50%MBM	0%MBM+G	25%MBM+G	50%MBM+G	
Arginine	8.15 a	8.33 a	7.20 b	8.43 a	8.11 a	7.63 b	0.31
Histidine	2.77	2.90	2.45	2.83	2.93	2.56	0.17
Isoleucine	4.65	4.97	4.18	4.19	4.74	4.36	0.12
Leucine	5.51	6.12	5.41	5.76	5.82	5.47	0.35
Lysine	6.64	6.54	6.28	6.97	6.32	6.28	0.32
Methionine	2.43	2.78	2.26	2.53	2.83	2.39	0.12
Phenylalanine	3.11	3.65	3.37	3.59	3.08	3.71	0.12
Threonine	3.64	3.76	3.81	3.73	3.83	3.88	0.25
Valine	3.13 ab	3.99 a	2.79 b	3.30 ab	3.73 ab	2.92 b	0.30
Non-essential amino acids							
Alanine	5.20	5.35	5.34	5.48	5.55	5.47	0.41
Aspartic acid	7.13 b	7.91 ab	8.32 ab	8.50 ab	8.03 ab	9.04 a	0.37
Glutamic acid	11.03	11.55	11.65	11.47	11.70	11.77	0.21
Glycine	5.96	5.77	5.45	5.46	5.65	5.29	0.41
Serine	3.06	3.92	3.40	3.91	3.41	3.07	0.22
Tyrosine	2.93	3.76	4.23	3.25	3.28	4.36	0.38
Σ EAA	40.03 ab	43.04 a	37.75 b	41.33 ab	41.39 ab	39.20 ab	1.36
Σ NEAA	35.41 b	38.19 a	38.39 a	38.03 ab	37.62 ab	39.00 a	1.37
Σ EAA/ Σ NEAA	1.13	1.12	0.98	1.08	1.10	1.00	0.22
TAA	75.44 b	81.23 a	76.34 b	78.60 a	79.01 a	78.20 ab	1.65

Note: Dietary treatments with different levels of fish meal (FM) replacement with meat and bone meal (MBM) with (G) or without (WG) garlic powder: 0%MBM (control without garlic powder), 25%MBM and 50%MBM (treatments without garlic powder; WG treatments). 0%MBM+G (control with garlic powder), 25%MBM+G and 50%MBM+G (treatments with garlic powder; G treatments).

Abbreviations: Σ EAA, sum of essential amino acids; Σ NEAA, sum of non-essential amino acids; TAAs, total amino acids.

to prevent decreasing growth. When garlic was supplemented to the diets, the level of replacement increased up to 50% (227 g/kg FM in diet) without decreasing growth performance. Therefore, the result of some studies who applied less than 300 g/kg FM as a control diet can be called into question (Arshad Hossain & Farzana Islam, 2007; Forster et al., 2003; Xie et al., 2018; Ye, Wang, Li, Sun, & Liu, 2011). However, some other researchers considered this point and thus formulated a diet for shrimp with higher than 400 g/kg FM as a control diet (Hernández et al., 2008; Tan et al., 2005). The outcomes of our work were compatible with other studies such as Tan et al. (2005) that reported 60% FM replacement by MBM (60% diet: 160 g/kg FM and 290 g/kg MBM) and also 35% replacing FM by MBM (35% diet: 258 g/kg FM and 204 g/kg MBM) was reported by Hernandez et al. (2008). Improvement of the digestion and absorption by garlic powder has been reported in whiteleg shrimp (Javadzadeh et al., 2012; Labrador et al., 2016) and fish such as sobaity sea bream (Jahanjoo et al., 2018), rainbow trout (Büyükdavacı et al., 2018; Esmaeili, Abedian Kenari, et al., 2017; Esmaeili, Kenari, et al., 2017), sterlet sturgeon (Lee et al., 2014), Asian sea bass (Talpur & Ikhwanuddin, 2012), Nile Tilapia (Naqi et al., 2019), common carp (Karimi Pashaki et al., 2018) and

goldfish (Dadgar et al., 2019). In some of these works, adding garlic to the diets could stimulate the digestive enzymes as well as increase the efficiency and possibly digestion and absorption of MBM in rainbow trout (Esmaeili, Kenari, et al., 2017). We could not analyse digestive enzymes in whiteleg shrimp, and further study is required. Another reason for the rise of growth performance in this study can be the increase in feed intake by adding garlic to their diet and in this way, shrimp fed 50%MBM+G had the highest consumed feed ration. An enhancement of feed intake with garlic in fish has widely been reported (Esmaeili, Kenari, et al., 2017; Guo et al., 2012; Talpur & Ikhwanuddin, 2012). Although weight gain was different, FCR in the present study was similar among treatments. Therefore, it can be stated that the increase in growth performance was due to increasing feed intake. Similar results have been reported in whiteleg shrimp (Xie et al., 2018; Ye et al., 2011).

In the present study, increasing inclusion of MBM in the diets decreased PER and LER values. Similarly, previous studies in rainbow trout (Bureau et al., 2000) indicated that a decrease in PER can be due to dietary deficiency of lysine, methionine and tryptophan. The decreasing PER in the present study can be due to this reason to some extent as 50%MBM and 50%MBM+G

TABLE 7 Effect of fish meal replacement by meat and bone meal on fatty acid profile of whiteleg shrimp muscle fed on various treatments after 60-day (g/100 g). Values are represented means $\pm$ standard deviations of triplicate samples. Means without letter are not significantly different. Letters a and b indicate significant differences in treatments, according to Duncan's multiple range tests ($p < .05$)

Fatty acids	Without garlic			With garlic			SDM
	0%MBM	25%MBM	50%MBM	0%MBM+G	25%MBM+G	50%MBM+G	
C14	0.76	0.81	0.65	0.76	0.45	0.66	0.24
C16	19.25	18.60	17.89	19.68	18.46	17.90	1.75
C18	8.98	9.21	9.14	9.00	8.44	9.10	0.54
C20	0.57	0.61	0.66	0.53	0.55	0.49	0.33
SFAs ^a	29.56 a	29.03 ab	28.34 ab	29.97 a	27.90 b	28.15 ab	1.85
C16:1n7	2.24	2.13	1.70	2.24	2.12	1.73	0.55
C18:1n9	20.88	20.70	20.30	20.95	20.77	20.37	1.77
MUFAs ^b	23.12	22.83	22.00	23.19	22.89	22.10	1.89
C18:2n6	10.24	10.95	10.37	11.35	10.96	11.00	1.02
C18:3n6	0.72 b	0.90 a	0.89 a	0.82 a	0.92 a	0.88 a	0.31
C20:4n6	3.25	3.16	3.41	3.27	3.13	3.39	0.11
C18:3n3	0.85	0.98	1.21	0.95	1.01	1.31	0.17
C20:5n-3	9.55	9.08	8.46	9.69	9.09	9.35	0.22
C22:6n-3	13.24 a	12.46 a	11.52 b	13.46 a	12.49 a	12.04 ab	0.19
PUFA ^c	37.85 ab	37.53 ab	35.86 b	39.54 a	37.59 ab	37.99 ab	1.86
$\Sigma n3^d$	23.64 a	22.52 ab	21.19 b	24.10 a	22.58 ab	22.70 ab	1.43
LC-PUFA ^e	26.22 a	24.81 ab	23.39 b	26.45 a	24.78 ab	24.90 ab	1.65
$\Sigma n6^f$	14.12	15.01	14.67	15.44	15.00	15.27	0.85
n3/n6	1.67	1.50	1.44	1.56	1.50	1.49	0.43

^aSaturated fatty acid—sum of all fatty acids without double bonds; includes 17:0 and 22:0 in addition to individually reported SFAs.

^bMonounsaturated fatty acids—sum of all fatty acids with a single double bond; includes 17:1n-7, 22:1n-11 and 22:1n-9 in addition to individually reported MUFAs.

^cPolyunsaturated fatty acids—sum of all fatty acids with ≥ 2 double bonds; includes 16:2n-4 in addition to individually reported PUFAs.

^dSum of all n-3 fatty acids.

^eLong-chain polyunsaturated fatty acids; sum of all C20 and C22 fatty acids with three or more double bonds; includes 20:3(n-6) in addition to other individually reported LC-PUFAs.

^fSum of all n-6 fatty acids; includes 20:3n-6 in addition to individually reported n-6 fatty acids. Dietary treatments with different levels of fish meal (FM) replacement with meat and bone meal (MBM) with (G) or without (WG) garlic powder: 0%MBM (control without garlic powder), 25%MBM and 50%MBM (treatments without garlic powder; WG treatments). 0%MBM+G (control with garlic powder), 25%MBM+G and 50%MBM+G (treatments with garlic powder; G treatments).

diets showed a deficiency in threonine, phenylalanine and methionine.

Although there was no significant difference in survival rate among groups, whiteleg shrimp fed garlic supplementation had higher content which is consistent with other studies (Dadgar et al., 2019; Esmaeili, Kenari, et al., 2017; Naqi et al., 2019; Talpur & Ikhwanuddin, 2012). MBM had no effect on survival rate in our research that is compatible with other works (Kureshy, Davis, & Arnold, 2000) and inconsistent with those reported decreasing survival ratio with replacing FM by MBM (Esmaeili, Abedian Kenari, et al., 2017).

4.2 | Muscle composition

Although differences in the chemical composition in aquatic species depend on internal factors such as age, gender, size and external factors

such as water quality, season and geographical area; the main reason usually comes from the diet (Shearer, 1994). In the present study, there was no significant difference in the protein and lipid contents of the whiteleg shrimp fed different experimental diets. However, with increased levels of replacement, fat and ash contents subsequently increased (Table 6), which was consistent with other studies (Ai et al., 2006; Ye et al., 2011). Furthermore, previous studies showed that body composition in aquatic animals did not change with replacing FM with animal proteins sources (Hu et al., 2008). Increasing level of MBM in the diets decreased ash content in whiteleg shrimp muscle which is incompatible with other works who reported increasing ash content in muscle with increasing MBM in the diets (Ye et al., 2011) and also in parallel with some studies (Qiu & Davis, 2018). It is reported the ability of using different protein sources is probably related to some intermediary metabolism so that the efficiency of protein synthesis and degradation can play important role (De la Higuera et al., 1999).

Furthermore, poor protein and amino acid utilization (i.e., efficiency and retention) resulted in somatic deposition of fat and decreasing protein in body (Esmaili, Kenari, et al., 2017). When we noticed amino acid profile in whiteleg shrimp muscle, we can confirm this hypothesis that amino acid utilization was in optimum level and therefore, protein and lipid contents did not change in whiteleg shrimp muscle with the inclusion of MBM in the diets. Another reason for stability of protein and fat contents in muscle can be due to the antioxidant activity of garlic. Garlic reduced the oxidation of fat and protein in muscle (Esmaili, Abedian Kenari, et al., 2017; Naeiji, Shahsavani, & Baghshani, 2013).

4.3 | Amino acids

For optimum feed formulation, amino acid composition of the diet should be very similar to amino acid profile in aquatic animal body (Wilson, 2003). Amino acid in diet should provide the amino acid requirement of animals for protein synthesis and also energy substrates (Rønnestad & Fyhn, 1993). In the present study, threonine, phenylalanine and methionine in 50%MBM and 50%MBM+G were lower than the requirements of whiteleg shrimp for these amino acids. However, whiteleg shrimp could utilize amino acids properly and in muscle, there were no significant signs of deficiency of these amino acids. The only deficiency in arginine was observed so that individuals fed 50%MBM and 50%MBM+G diets had significantly lower value of this amino acid as compared to other group which was parallel to the other study (Ye et al., 2011). One reason for the lack of growth impairment in whiteleg shrimp fed 50%MBM+G can be due to the fact that this species could utilize amino acid optimally thanks to garlic supplementation. The increase in free amino acid contents in liver and muscle of catfish (*Clarias lazera*) by garlic and onion supplementation was reported (Al-Salahy, 2002). One possible reason for this could be the oxidation of free or combined thiol compounds in a protein molecule by the available sulphur components in the onion and garlic (Corzo-Martínez, Corzo, & Villamiel, 2007). Esmaili, Abedian Kenari, et al. (2017) also reported improved amino acid utilization in rainbow trout from adding garlic. Kureshy et al. (2000) reported methionine is the first limiting amino acid in MBM and because of this, we supplemented diets with methionine (5 g/kg diet) and it could provide the methionine requirement of whiteleg shrimp (1.4 g/100 g) with up to 25% replacement. However, it is recommended to add more methionine in diet with high levels of replacement to satisfy the requirements for whiteleg shrimp. A possible reason for the decreasing growth in whiteleg shrimp fed 50%MBM could be the shortage of this amino acid in their diet as Tan et al. (2005) proposed this phenomena as well. Although the body synthesizes non-essential amino acids (NEAAs) sufficiently, there has been no strong experimental evidence to support this hypothesis over the past century. NEAAs play key roles in regulating gene expression, cell signalling, antioxidative responses, neurotransmission and immunity. Furthermore, glutamate, glutamine and aspartate are major metabolic fuels for the small intestine to maintain its digestive function and protect its mucosal integrity (Wu et al.,

2013). In the present study, the amount of aspartic acid and glutamic acid in treatments with garlic was higher and it may be a reason for high growth performance in whiteleg shrimp fed 50%MBM+G diet. In this sense, Xie et al. (2018) reported low protein utilization efficiency in whiteleg shrimp fed diets with low amount of FM and also alanine, aspartate and glutamate metabolism pathways were down-regulated in individuals fed those diets. While lower digestibility of many amino acids of MBM for whiteleg shrimp has been reported (Liu, Ye, Kong, Wang, & Wang, 2013), our study did not show any supporting evidence for this phenomenon.

4.4 | Fatty acid profiles

Fatty acid composition in body reflects the fatty acid profile of the diet (Matani Bour, Esmaili, & Abedian Kenari, 2018; Roohani et al., 2019; Turchini, Torstensen, & Ng, 2009). In the present study, increasing MBM in the diets decreased $\Sigma n3$ and PUFA values and increased MUFA and C18:2n6 levels. However, these changes in whiteleg shrimp muscle were not similarly observed so that just minor decreases in PUFA and $\Sigma n3$ were found. However, dissimilar results have been reported in fish studies and increasing levels of SFAs and MUFAs in diets with high levels of MBM coincided with the increasing of these fatty acids in muscle (Ai et al., 2006; Esmaili, Abedian Kenari, et al., 2017; Millamena & Golez, 2001). It has been reported that aquatic animals can elongate and desaturate the C18 precursors into the physiologically important LC-PUFAs (Hastings et al., 2001) and this process in whiteleg shrimp fed 0%MBM+G group probably occurred as the amount of PUFA and $\Sigma n3$ in muscle was higher than other groups. To confirm this issue, a competition between 18:2n-6 and 18:3n-3, as both fatty acids are substrates for biosynthesis of PUFAs, was observed (de Antueno et al., 2001). However, $\Sigma n6$ in the present study was similar among shrimps and we can conclude that shrimp has different pathways to provide PUFAs.

Decreasing growth performance of whiteleg shrimp that were fed a diet with high amounts of MBM was reported (Ye et al., 2011), which was in parallel with decreasing growth in shrimp fed 50%MBM diet in our work. Many studies have also shown that unsaturated fatty acids are essential for maximizing growth and optimizing the feeding of whiteleg shrimp (González-Félix, Gatlin, Lawrence, & Perez-Velazquez, 2002; Lim, Ako, Brown, & Hahn, 1997; Luo et al., 2018; Soller, Roy, & Davis, 2019). In our study, dietary fatty acids especially $\Sigma n3$ were well digested and utilized in whiteleg shrimp fed garlic supplementation. Therefore, accumulation of EPA and DHA in the whiteleg shrimp muscle is likely associated with antioxidant properties of garlic (Chung, 2006) which could protect $\Sigma n3$ fatty acids from oxidation.

5 | CONCLUSION

The results of the present study provide sufficient evidence that whiteleg shrimp can use the 50%MBM+G (227 g/kg FM

and 303 g/kg MBM plus 30 g/kg garlic) diet without decreasing growth performance, impairing muscle quality, amino acid and fatty acid profiles. We hypothesized garlic could increase the level of replacement by three actions: 1. increasing feed intake, 2. improving protein and amino acid utilization, and 3. protecting fatty acids from oxidation. As the most available animal by-product, 50%MBM+G can be recommended for ensuring the sustainability of shrimp aquaculture by decreasing the cost of diet and pressure of using FM.

ACKNOWLEDGMENTS

The authors wish to thank the Tarbiat Modares University (Noor, Mazandaran, Iran) and Center of Reproduction and Cultivation of shrimp (Gomishan, Golestan, Iran) for their financial and technical support. Thanks are also extended to Morteza Kamali for assisting our analytical experiments.

CONFLICT OF INTEREST

There is no conflict of interest in this paper.

ETHICAL STATEMENT

All applicable national and institutional guidelines for the care and use of shrimp were according to the Iran Veterinary Organization and also Tarbiat Modares University. (The guidelines adopted from the Declaration of Helsinki (1975) and the Society for Neuroscience Animal Care and Use Guidelines (1998) were approved for implementation by the Medical Ethics Committee, School of Medical Sciences of the Tarbiat Modares University on 28 Farvardin 1385/17 April 2006.)

ORCID

Abdolmohammad Abedian Kenari  <https://orcid.org/0000-0003-3410-3908>

DATA AVAILABILITY STATEMENT

The data cannot be shared before publishing.

REFERENCES

- Abtahi, B., Yousefi, M., & Kenari, A. A. (2013). Influence of dietary nucleotides supplementation on growth, body composition and fatty acid profile of Beluga sturgeon juveniles (*Huso huso*). *Aquaculture Research*, 44(2), 254–260.
- Ai, Q., Mai, K., Tan, B., Xu, W., Duan, Q., Ma, H., & Zhang, L. (2006). Replacement of fish meal by meat and bone meal in diets for large yellow croaker, *Pseudosciaena crocea*. *Aquaculture*, 260(1–4), 255–263. <https://doi.org/10.1016/j.aquaculture.2006.06.043>
- Akiyama, D. M. (1992). Penaeid shrimp nutrition. In A. W. Fast & L. J. Lester (Eds.), *Marine shrimp culture: Principles and practice* (pp. 535–567). Elsevier Science. Retrieved from <https://www.elsevier.com/books/marine-shrimp-culture/fast/978-0-444-88606-4>
- Aksnes, A., & Opstvedt, J. (1998). Content of digestible energy in fish feed ingredients determined by the ingredient-substitution method. *Aquaculture*, 161(1–4), 45–53. [https://doi.org/10.1016/S0044-8486\(97\)00255-X](https://doi.org/10.1016/S0044-8486(97)00255-X)
- Al-Salahy, M. (2002). Some physiological studies on the effect of onion and garlic juices on the fish, *Clarias lazera*. *Fish Physiology and Biochemistry*, 27(1–2), 129–142. <https://doi.org/10.1023/B:FISH.0000021913.60189.76>
- AOAC (1995). AOAC (Association of Official Analytical Chemists). In P. Cunniff (Ed.), *Official methods of analysis* (16th ed.), Arlington, VA. Retrieved from <https://www.worldcat.org/title/official-methods-of-analysis-of-aoac-international/oclc/421897987>
- Arshad Hossain, M., & Farzana Islam, S. (2007). Meat and bone meal as partial substitute for fish meal in nursery diet for giant freshwater prawn, *Macrobrachium rosenbergii* (De Man). *Journal of the World Aquaculture Society*, 38(2), 272–280. <https://doi.org/10.1111/j.1749-7345.2007.00097.x>
- Bureau, D., Harris, A., Bevan, D., Simmons, L., Azevedo, P., & Cho, C. (2000). Feather meals and meat and bone meals from different origins as protein sources in rainbow trout (*Oncorhynchus mykiss*) diets. *Aquaculture*, 181(3), 281–291. [https://doi.org/10.1016/S0044-8486\(99\)00232-X](https://doi.org/10.1016/S0044-8486(99)00232-X)
- Bureau, D., Harris, A., & Cho, C. (1999). Apparent digestibility of rendered animal protein ingredients for rainbow trout (*Oncorhynchus mykiss*). *Aquaculture*, 180(3), 345–358. [https://doi.org/10.1016/S0044-8486\(99\)00210-0](https://doi.org/10.1016/S0044-8486(99)00210-0)
- Büyükdavacı, M. E., Balcázar, J. L., Demirkale, İ., & Dikel, S. (2018). Effects of garlic-supplemented diet on growth performance and intestinal microbiota of rainbow trout (*Oncorhynchus mykiss*). *Aquaculture*, 486, 170–174. <https://doi.org/10.1016/j.aquaculture.2017.12.022>
- Cao, L., Naylor, R., Henriksson, P., Leadbitter, D., Metian, M., Troell, M., & Zhang, W. (2015). China's aquaculture and the world's wild fisheries. *Science*, 347(6218), 133–135.
- Chung, L. Y. (2006). The antioxidant properties of garlic compounds: Allyl cysteine, alliin, allicin, and allyl disulfide. *Journal of Medicinal Food*, 9(2), 205–213.
- Corzo-Martínez, M., Corzo, N., & Villamiel, M. (2007). Biological properties of onions and garlic. *Trends in Food Science & Technology*, 18(12), 609–625.
- Dadgar, S., Seidgar, M., Nekuiefard, A., Valipour, A., Sharifian, M., & Hafezieh, M. (2019). Oral administration of garlic powder (*Allium sativum*) on growth performance and survival rate of *Carassius auratus* fingerlings. *Iranian Journal of Fisheries Sciences*, 18(1), 71–82.
- de Antueno, R. J., Knickle, L. C., Smith, H., Elliot, M. L., Allen, S. J., Nwaka, S., & Winther, M. D. (2001). Activity of human $\Delta 5$ and $\Delta 6$ desaturases on multiple n-3 and n-6 polyunsaturated fatty acids. *FEBS Letters*, 509(1), 77–80.
- De la Higuera, M., Akharbach, H., Hidalgo, M., Peragon, J., Lupianez, J., & Garcia-Gallego, M. (1999). Liver and white muscle protein turnover rates in the European eel (*Anguilla anguilla*): Effects of dietary protein quality. *Aquaculture*, 179(1–4), 203–216.
- Esmaili, M., Abedian Kenari, A., & Rombenso, A. (2017). Effects of fish meal replacement with meat and bone meal using garlic (*Allium sativum*) powder on growth, feeding, digestive enzymes and apparent digestibility of nutrients and fatty acids in juvenile rainbow trout (*Oncorhynchus mykiss*). *Aquaculture Nutrition*, 23(6), 1225–1234.
- Esmaili, M., Kenari, A. A., & Rombenso, A. (2017). Immunohematological status under acute ammonia stress of juvenile rainbow trout (*Oncorhynchus mykiss* Walbaum, 1792) fed garlic (*Allium sativum*) powder-supplemented meat and bone meal-based feeds. *Comparative Clinical Pathology*, 26(4), 853–866.
- FAO (2018). *The state of world fisheries and aquaculture 2018*. Rome: Food and Agriculture Organization of the United Nations.
- Firestone, D. (2009). Official methods and recommended practices of the AOCS. Urbana, IL: AOCS.
- Folch, J., Lees, M., & Sloane-Stanley, G. (1957). A simple method for the isolation and purification of total lipids from animal tissues. *Journal of Biological Chemistry*, 226(1), 497–509.
- Forster, I., Dominy, W., Obaldo, L., & Tacon, A. (2003). Rendered meat and bone meals as ingredients of diets for shrimp *Litopenaeus*

- vannamei (Boone, 1931). *Aquaculture*, 219(1–4), 655–670. [https://doi.org/10.1016/S0044-8486\(02\)00457-X](https://doi.org/10.1016/S0044-8486(02)00457-X)
- González-Félix, M. L., Gatlin, D. M. III, Lawrence, A. L., & Perez-Velazquez, M. (2002). Effect of dietary phospholipid on essential fatty acid requirements and tissue lipid composition of *Litopenaeus vannamei* juveniles. *Aquaculture*, 207(1–2), 151–167. [https://doi.org/10.1016/S0044-8486\(01\)00797-9](https://doi.org/10.1016/S0044-8486(01)00797-9)
- Guo, J., Kuo, C., Chuang, Y., Hong, J., Chou, R., & Chen, T. (2012). The effects of garlic-supplemented diets on antibacterial activity against *Streptococcus iniae* and on growth in orange-spotted grouper, *Epinephelus coioides*. *Aquaculture*, 364, 33–38. <https://doi.org/10.1016/j.aquaculture.2012.07.023>
- Hastings, N., Agaba, M., Tocher, D. R., Leaver, M. J., Dick, J. R., Sargent, J. R., & Teale, A. J. (2001). A vertebrate fatty acid desaturase with $\Delta 5$ and $\Delta 6$ activities. *Proceedings of the National Academy of Sciences*, 98(25), 14304–14309.
- Hernández, C., Olvera-Novoa, M. A., Aguilar-Vejar, K., González-Rodríguez, B., & de la Parra, I. A. (2008). Partial replacement of fish meal by porcine meat meal in practical diets for Pacific white shrimp (*Litopenaeus vannamei*). *Aquaculture*, 277(3–4), 244–250. <https://doi.org/10.1016/j.aquaculture.2008.02.016>
- Hosseinpour Aghaei, R., Abedian Kenari, A., Yazdani Sadati, M. A., & Esmaeili, M. (2018). The effect of time-dependent protein restriction on growth factors, nonspecific immunity, body composition, fatty acids and amino acids in the Siberian sturgeon (*Acipenser baerii*). *Aquaculture Research*, 49(9), 3033–3044.
- Hu, M., Wang, Y., Luo, Z., Zhao, M., Xiong, B., Qian, X., & Zhao, Y. (2008). Evaluation of rendered animal protein ingredients for replacement of fish meal in practical diets for gibel carp, *Carassius auratus gibelio* (Bloch). *Aquaculture Research*, 39(14), 1475–1482.
- Jahanjoo, V., Yahyavi, M., Akrami, R., & Bahri, A. H. (2018). Influence of adding garlic (*Allium sativum*), Ginger (*Zingiber officinale*), thyme (*Thymus vulgaris*) and their combination on the growth performance, haematoimmunological parameters and disease resistance to *Photobacterium damsela* in sobaity sea bream (*Sparidentex hasta*) Fry. *Turkish Journal of Fisheries and Aquatic Sciences*, 18(4), 633–645. https://doi.org/10.4194/1303-2712-v18_4_15
- Javadzadeh, M., Salarzadeh, A., Yahyavi, M., Hafezieh, M., & Darvishpour, H. (2012). Effect of garlic extract on growth and survival rate in *Litopenaeus vannamei* post larvae. *Iranian Scientific Fisheries Journal*, 21(1), 39–46.
- Karimi Pashaki, A., Zorriehzahra*, S. J., Ghasemi, M., Sharif Rohani, M., & Hosseini, S. M. (2018). Effects of dietary Garlic extract on some blood, immunity and growth parameters of Common Carp fingerlings (*Cyprinus carpio*). *Iranian Journal of Aquatic Animal Health*, 4(2), 28–39. <https://doi.org/10.29252/ijaah.4.2.28>
- Kenari, A. A., Sotoudeh, E., & Rezaei, M. H. (2011). Dietary soybean phosphatidylcholine affects growth performance and lipolytic enzyme activity in Caspian brown trout (*Salmo trutta Caspius*) alevin. *Aquaculture Research*, 42(5), 655–663. <https://doi.org/10.1111/j.1365-2109.2010.02587.x>
- Khalil, R., Nadia, B., & Soliman, M. (2001). Effects of Biogen and Levamisol Hcl on the immune response of cultured *Oreochromis niloticus* to *Aeromonas hydrophila* vaccine. *Beni-Suif Veterinary Medicine Journal Egypt XI*, 2, 381–392.
- Kureshy, N., Davis, D. A., & Arnold, C. (2000). Partial replacement of fish meal with meat-and-bone meal, flash-dried poultry by-product meal, and enzyme-digested poultry by-product meal in practical diets for juvenile red drum. *North American Journal of Aquaculture*, 62(4), 266–272. [https://doi.org/10.1577/1548-8454\(2000\)062<0266:PROFMW>2.0.CO;2](https://doi.org/10.1577/1548-8454(2000)062<0266:PROFMW>2.0.CO;2)
- Labrador, J. R. P., Guiñares, R. C., & Hontiveros, G. J. S. (2016). Effect of garlic powder-supplemented diets on the growth and survival of Pacific white leg shrimp (*Litopenaeus vannamei*). *Cogent Food & Agriculture*, 2(1), 1210066. <https://doi.org/10.1080/2331932.2016.1210066>
- Lee, D.-H., Lim, S.-R., Han, J.-J., Lee, S.-W., Ra, C.-S., & Kim, J.-D. (2014). Effects of dietary garlic powder on growth, feed utilization and whole body composition changes in fingerling sterlet sturgeon, *Acipenser ruthenus*. *Asian-Australasian Journal of Animal Sciences*, 27(9), 1303–1310. <https://doi.org/10.5713/ajas.2014.14087>
- Lee, J. Y., & Gao, Y. (2012). Review of the Application of Garlic, *Allium sativum*, in Aquaculture. *Journal of the World Aquaculture Society*, 43(4), 447–458. <https://doi.org/10.1111/j.1749-7345.2012.00581.x>
- Lim, C., Ako, H., Brown, C. L., & Hahn, K. (1997). Growth response and fatty acid composition of juvenile *Penaeus vannamei* fed different sources of dietary lipid. *Aquaculture*, 151(1–4), 143–153. [https://doi.org/10.1016/S0044-8486\(96\)01500-1](https://doi.org/10.1016/S0044-8486(96)01500-1)
- Liu, X.-H., Ye, J.-D., Kong, J.-H., Wang, K., & Wang, A.-L. (2013). Apparent digestibility of 12 protein-origin ingredients for Pacific white shrimp *Litopenaeus vannamei*. *North American Journal of Aquaculture*, 75(1), 90–98.
- Luo, N., Ding, Z. L., Kong, Y. Q., Zhang, R. F., Zhang, Y. X., Wu, C. L., ... Ye, J. Y. (2018). An evaluation of increasing linolenic acid level in the diet of *Macrobrachium nipponense*: Lipid deposition, fatty acid composition and expression of lipid metabolism-related genes. *Aquaculture Nutrition*, 24(2), 758–767.
- Matani Bour, H., Esmaeili, M., & Abedian Kenari, A. (2018). Growth performance, muscle and liver composition, blood traits, digestibility and gut bacteria of beluga (*Huso huso*) juvenile fed different levels of soybean meal and lactic acid. *Aquaculture Nutrition*, 24(4), 1361–1368.
- Meeker, D. L. (2006). *Essential rendering: All about the animal by-products industry*. National Renderers Association. Retrieved from <https://www.amazon.com/Essential-Rendering-Animal-Products-Industry/dp/09654466035>
- Millamena, O., & Golez, N. (2001). Evaluation of processed meat solubles as replacement for fish meal in diet for juvenile grouper *Epinephelus coioides* (Hamilton). *Aquaculture Research*, 32(s1), 281–287.
- Naeiji, N., Shahsavani, D., & Baghshani, H. (2013). Effect of dietary garlic supplementation on lipid peroxidation and protein oxidation biomarkers of tissues as well as some serum biochemical parameters in common carp *Cyprinus carpio*. *Fisheries Science*, 79(4), 699–705.
- Naqi, J. A., Mateen, A., Hussain, D., Tahir, H. M., Hussain, S., & Tabasum, A. (2019). Effect of *Allium sativum* supplemented diets on growth and haematological responses in Nile Tilapia (*Oreochromis niloticus*). *Pakistan Journal of Zoology*, 51(1), 257–263.
- NRC (National Research Council) (2011). *Nutrient requirements of fish and shrimp*. Washington DC: National Academies Press.
- Olsen, R. L., & Hasan, M. R. (2012). A limited supply of fishmeal: Impact on future increases in global aquaculture production. *Trends in Food Science & Technology*, 27(2), 120–128.
- Oujifard, A., Seyfabadi, J., Kenari, A. A., & Rezaei, M. (2012). Fish meal replacement with rice protein concentrate in a practical diet for the Pacific white shrimp, *Litopenaeus vannamei* Boone, 1931. *Aquaculture International*, 20(1), 117–129.
- Platel, K., & Srinivasan, K. (2004). Digestive stimulant action of spices: A myth or reality? *The Indian Journal of Medical Research*, 119(5), 167–179.
- Qiu, X., & Davis, D. A. (2018). Use of porcine meal in plant-based practical diets for pacific White Shrimp, *Litopenaeus vannamei*. *Journal of the World Aquaculture Society*, 49(3), 582–589.
- Ramezanzadeh, S., Abedian Kenari, A., & Esmaeili, M. (2019). Immunohematological parameters of rainbow trout (*Oncorhynchus mykiss*) fed supplemented diet with different forms of barberry root (*Berberis vulgaris*). *Comparative Clinical Pathology*, 1–11. <https://doi.org/10.1007/s00580-019-03032-8>
- Ramezanzadeh, S., Abedian Kenari, A., Esmaeili, M., & Rombenso, A. (2019). Effects of different forms of barberry root (*Berberis vulgaris*)

- on growth performance, muscle fatty acids profile, whole body composition and digestive enzymes of rainbow trout (*Oncorhynchus mykiss*). *Journal of the World Aquaculture Society*.
- Rønnestad, I., & Fyhn, H. J. (1993). Metabolic aspects of free amino acids in developing marine fish eggs and larvae. *Reviews in Fisheries Science*, 1(3), 239–259. <https://doi.org/10.1080/10641269309388544>
- Roohani, A. M., Abedian Kenari, A., Fallahi Kapoorchali, M., Borani, M. S., Zoriezahra, S. J., Smiley, A. H., ... Rombenso, A. N. (2019). Effect of spirulina *Spirulina platensis* as a complementary ingredient to reduce dietary fish meal on the growth performance, whole-body composition, fatty acid and amino acid profiles, and pigmentation of Caspian brown trout (*Salmo trutta caspius*) juveniles. *Aquaculture Nutrition*, 25(3), 633–645.
- Safavi, S. V., Abedian Kenari, A., Tabarsa, M., & Esmaeili, M. (2019). Effect of sulfated polysaccharides extracted from marine macroalgae (*Ulva intestinalis* and *Gracilariopsis persica*) on growth performance, fatty acid profile, and immune response of rainbow trout (*Oncorhynchus mykiss*). *Journal of Applied Phycology*, 1–15. <https://doi.org/10.1007/s10811-019-01902-w>
- Shearer, K. D. (1994). Factors affecting the proximate composition of cultured fishes with emphasis on salmonids. *Aquaculture*, 119(1), 63–88. [https://doi.org/10.1016/0044-8486\(94\)90444-8](https://doi.org/10.1016/0044-8486(94)90444-8)
- Soller, F., Roy, L. A., & Davis, D. A. (2019). Replacement of fish oil in plant-based diets for Pacific white shrimp, *Litopenaeus vannamei*, by stearine fish oil and palm oil. *Journal of the World Aquaculture Society*, 50(1), 186–203.
- Sookying, D., Davis, D., Silva, S. D. D., & F., (2013). A review of the development and application of soybean-based diets for Pacific white shrimp *Litopenaeus vannamei*. *Aquaculture Nutrition*, 19(4), 441–448.
- Talpur, A. D., & Ikhwanuddin, M. (2012). Dietary effects of garlic (*Allium sativum*) on haemato-immunological parameters, survival, growth, and disease resistance against *Vibrio harveyi* infection in Asian sea bass, *Lates calcarifer* Bloch. *Aquaculture*, 364, 6–12. <https://doi.org/10.1016/j.aquaculture.2012.07.035>
- Tan, B., Mai, K., Zheng, S., Zhou, Q., Liu, L., & Yu, Y. (2005). Replacement of fish meal by meat and bone meal in practical diets for the white shrimp *Litopenaeus vannamei* (Boone). *Aquaculture Research*, 36(5), 439–444. <https://doi.org/10.1111/j.1365-2109.2005.01223.x>
- Turchini, G. M., Torstensen, B. E., & Ng, W. K. (2009). Fish oil replacement in finfish nutrition. *Reviews in Aquaculture*, 1(1), 10–57. <https://doi.org/10.1111/j.1753-5131.2008.01001.x>
- Turchini, G. M., Trushenski, J. T., & Glencross, B. D. (2019). Thoughts for the future of aquaculture nutrition: Realigning perspectives to reflect contemporary issues related to judicious use of marine resources in aquafeeds. *North American Journal of Aquaculture*, 81(1), 13–39. <https://doi.org/10.1002/naaq.10067>
- Webster, C. D., Tiu, L. G., & Tidwell, J. H. (1997). Effects of replacing fish meal in diets on growth and body composition of palmetto bass (*Morone saxatilis* × *M. chrysops*) raised in cages. *Journal of Applied Aquaculture*, 7(1), 53–67.
- Wilson, R. P. (2003). Amino acids and proteins. In *Fish nutrition* (pp. 143–179). Elsevier. Retrieved from <https://www.elsevier.com/books/fish-nutrition/unknown/978-0-12-319652-1>
- Wu, G., Wu, Z., Dai, Z., Yang, Y., Wang, W., Liu, C., ... Yin, Y. (2013). Dietary requirements of "nutritionally non-essential amino acids" by animals and humans. *Amino Acids*, 44(4), 1107–1113. <https://doi.org/10.1007/s00726-012-1444-2>
- Xie, S., Niu, J., Zhou, W., Liu, Y., & Tian, L. (2018). Developing a low fish-meal diet for juvenile Pacific white shrimp, *Litopenaeus vannamei*, using the nutritional value of FM as the reference profile. *Aquaculture Nutrition*, 24(4), 1184–1197.
- Ye, J. D., Wang, K., Li, F. D., Sun, Y. Z., & Liu, X. H. (2011). Incorporation of a mixture of meat and bone meal, poultry by-product meal, blood meal and corn gluten meal as a replacement for fish meal in practical diets of Pacific white shrimp *Litopenaeus vannamei* at two dietary protein levels. *Aquaculture Nutrition*, 17(2), e337–e347. <https://doi.org/10.1111/j.1365-2095.2010.00768.x>
- Zeilab Sendijani, R., Abedian Kenari, A., Smiley, A. H., & Esmaeili, M.. (2019). The effect of extract from dill (*Anethum graveolens*) on the growth performance, body composition, immune system and antioxidant system of Rainbow Trout (*Oncorhynchus mykiss*). *North American Journal of Aquaculture* (in press). <https://doi.org/10.1002/naaq.10123>

How to cite this article: Tazikeh T, Abedian Kenari A, Esmaeili N. Effects of fish meal replacement by meat and bone meal supplemented with garlic (*Allium sativum*) powder on biological indices, feeding, muscle composition, fatty acid and amino acid profiles of whiteleg shrimp (*Litopenaeus vannamei*). *Aquac Res*. 2020;51:674–686. <https://doi.org/10.1111/are.14416>